METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

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LINE SIZING LIST

Process lines of P&IDs MEOH-PID-001 and MEOH-PID-002 rev A

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DOCUMENT TITLE : Line Sizing List

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Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

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1 Purpose and scope

One row per numbered process line on the EPC P&IDs MEOH-PID-001 and MEOH-PID-002 rev A, with the line number and nominal size as drawn. Flow, temperature, pressure and density come from the heat and material balance MEOH-V1-PE-HMB-001 (simulator model); the check is against typical sizing criteria quoted as typical practice (assumed). Where the drawn size does not meet the criteria at the rated flow, the size that would is given and an engineering query is raised.

2 Basis

Service	Criteria (typical practice, assumed)
Gas, high pressure (PG, SG, PU upstream of PV-203)	Gas, continuous, P > 50 bar g: v ≤ 20 m/s (the P&ID note 2 basis), ρv² ≤ 15 000 kg/(m·s²) (typical noise / vibration practice, assumed), ΔP ≤ 0.5 bar/100 m (typical for high-pressure gas, assumed)
Gas, low pressure (PU downstream of PV-203)	Gas, low pressure (fuel header side): v ≤ 30 m/s, ρv² ≤ 15 000, ΔP ≤ 0.5 bar/100 m (typical, assumed)
Liquid (CM upstream of LV-204)	Liquid, single phase: v ≤ 2 m/s (the P&ID note 2 basis), ΔP ≤ 0.9 bar/100 m (typical, assumed)

Rated flow = the larger of normal × 1.10 and the highest flow of the four HMB cases, each at its own density (the governing case is named in the list). ρv² and ΔP are evaluated at the rated flow. ΔP per 100 m of straight pipe, Darcy friction factor from the Churchill (1977) equation, roughness 0.046 mm (assumed). Gas density ideal gas (Z = 1) as in the model.

2.1 Design conditions and pipe inside diameters

Design pressure and temperature of each line from the equipment list MEOH-V1-ME-EQL-001: zone A 88 bar g (V-201, C-201 suction; PSV-201 set), zone B 100 bar g (C-201 discharge to the V-201 inlet), design temperature of the equipment the line connects. Pipe material carbon steel A106 Gr B up to 220 °C (the EQL carbon-steel limit in hot hydrogen), 1¼Cr-½Mo A335 P11 above; flanges Class 600 zone A, Class 900 zone B (assumed).

DN	OD, mm	Sch 80, mm	B 100 bar g / 170 °C CS S 138	B 100 bar g / 335 °C 1¼Cr-½Mo S 125	A 88 bar g / 120 °C CS S 138	B 100 bar g / 120 °C CS S 138	B 100 bar g / 280 °C 1¼Cr-½Mo S 125	B 100 bar g / 225 °C 1¼Cr-½Mo S 125
150	168.3	10.97	t 11.0 · ID 146.4	t 11.0 · ID 146.4	t 11.0 · ID 146.4	t 11.0 · ID 146.4	t 11.0 · ID 146.4	t 11.0 · ID 146.4
200	219.1	12.70	t 12.7 · ID 193.7	t 12.7 · ID 193.7	t 12.7 · ID 193.7	t 12.7 · ID 193.7	t 12.7 · ID 193.7	t 12.7 · ID 193.7
300	323.9	17.48	t 17.5 · ID 288.9	t 17.5 · ID 288.9	t 17.5 · ID 288.9	t 17.5 · ID 288.9	t 17.5 · ID 288.9	t 17.5 · ID 288.9
350	355.6	19.05	t 19.1 · ID 317.5	t 19.1 · ID 317.5	t 19.1 · ID 317.5	t 19.1 · ID 317.5	t 19.1 · ID 317.5	t 19.1 · ID 317.5
400	406.4	21.44	t 21.4 · ID 363.5	t 21.4 · ID 363.5	t 21.4 · ID 363.5	t 21.4 · ID 363.5	t 21.4 · ID 363.5	t 21.4 · ID 363.5
600	610.0	30.96	t 31.0 · ID 548.1	t 31.0 · ID 548.1	t 31.0 · ID 548.1	t 31.0 · ID 548.1	t 31.0 · ID 548.1	t 31.0 · ID 548.1
650	660.0	—	t 29.0 · ID 602.0	t 31.0 · ID 598.0	t 26.0 · ID 608.0	t 29.0 · ID 602.0	t 31.0 · ID 598.0	t 31.0 · ID 598.0
750	762.0	—	t 33.0 · ID 696.0	t 36.0 · ID 690.0	t 29.0 · ID 704.0	t 33.0 · ID 696.0	t 36.0 · ID 690.0	t 36.0 · ID 690.0
800	813.0	—	t 35.0 · ID 743.0	t 38.0 · ID 737.0	t 31.0 · ID 751.0	t 35.0 · ID 743.0	t 38.0 · ID 737.0	t 38.0 · ID 737.0
900	914.0	—	t 39.0 · ID 836.0	t 43.0 · ID 828.0	t 35.0 · ID 844.0	t 39.0 · ID 836.0	t 43.0 · ID 828.0	t 43.0 · ID 828.0
1000	1,016.0	—	t 43.0 · ID 930.0	t 47.0 · ID 922.0	t 38.0 · ID 940.0	t 43.0 · ID 930.0	t 47.0 · ID 922.0	t 47.0 · ID 922.0
1050	1,067.0	—	t 45.0 · ID 977.0	t 49.0 · ID 969.0	t 40.0 · ID 987.0	t 45.0 · ID 977.0	t 49.0 · ID 969.0	t 49.0 · ID 969.0

Wall = Sch 80 where it exists and is enough, else the ASME B31.3 304.1.2 minimum t = P·D / (2(S·E + P·Y)) at the design pressure, E = 1, Y = 0.4, + 1.5 mm corrosion allowance, ÷ (1 − 0.125) mill tolerance, rounded up to 1 mm; S in MPa (A106 Gr B 138 MPa to 204 °C, 130 MPa to 260 °C; A335 P11 125 MPa to 345 °C; all assumed; the piping class confirms wall and ID).

3 Line sizing list

Line no. (as drawn)	Sheet	Service / route	Str.	P bar g	T °C	ρ kg/m³	Normal kg/h	Rated kg/h (governing)	Design (EQL zone)	ID mm	v normal m/s	v rated m/s	ρv² rated	ΔP rated bar/100 m	Status	Meets criteria at
DN750-PG-201	PID-001	Converter feed after E-202: E-202 cold side outlet (off-sheet) → Quench header / bed-1 feed branch	4	79.7	96.8	17.25	565,272	621,799 (normal × 1.10)	B: 100 bar g / 170 °C, CS Cl 900	696.0	23.93	26.32	11,948	0.097	outside: v	DN900
DN300-PG-222	PID-001	Cold shot to bed 2: Quench header → R-201 bed 2 distributor via TV-221	6/b2	79.7	96.8	17.25	71,674	78,841 (normal × 1.10)	B: 100 bar g / 170 °C, CS Cl 900	288.9	17.61	19.37	6,471	0.150	OK	drawn size
DN350-PG-232	PID-001	Cold shot to bed 3: Quench header → R-201 bed 3 distributor via TV-231	6/b3	79.7	96.8	17.25	115,293	126,822 (normal × 1.10)	B: 100 bar g / 170 °C, CS Cl 900	317.5	23.45	25.79	11,477	0.236	outside: v	DN400
DN350-PG-242	PID-001	Cold shot to bed 4: Quench header → R-201 bed 4 distributor via TV-241	6/b4	79.7	96.8	17.25	110,470	124,602 (Max rate)	B: 100 bar g / 170 °C, CS Cl 900	317.5	22.47	24.85	10,865	0.223	outside: v	DN400
DN600-PG-213	PID-001	Bed 1 feed (after E-201, E-204): E-204 outlet → R-201 top nozzle	5	79.7	220.0	12.94	267,836	380,941 (End of run)	B: 100 bar g / 335 °C, 1¼Cr-½Mo Cl 900	548.1	24.37	31.72	14,225	0.153	outside: v	DN800
DN900-PG-251	PID-001	Converter effluent: R-201 bottom (bed 4 outlet) → E-201 hot side inlet	7	77.4	267.6	12.48	565,264	621,790 (normal × 1.10)	B: 100 bar g / 335 °C, 1¼Cr-½Mo Cl 900	828.0	23.37	25.70	8,245	0.055	outside: v	DN1050
DN900-PG-205	PID-002	V-201 vapour / recycle gas: V-201 top → C-201 suction (recycle; purge tee on the riser)	10	75.0	40.0	17.02	433,748	477,123 (normal × 1.10)	A: 88 bar g / 120 °C, CS Cl 600	844.0	12.65	13.92	3,297	0.021	OK	drawn size
DN150-PU-203	PID-002	Purge gas: V-201 vapour riser (tee) → Reformer fuel header via PV-203, FI-203	11	75.0	40.0	17.02	27,337	37,875 (End of run)	A: 88 bar g / 120 °C, CS Cl 600	146.4	26.50	31.60	19,748	1.027	outside: v, ρv², ΔP	DN200
DN200-CM-209	PID-002	Crude methanol: V-201 bottom → Let-down / distillation via LV-204, FI-209	12	75.0	40.0	814	131,516	144,668 (normal × 1.10)	A: 88 bar g / 120 °C, CS Cl 600	193.7	1.52	1.67	2,284	0.094	OK	drawn size
DN300-SG-202	PID-002	Make-up syngas: Battery limit (syngas compressor) → C-201 suction header via FV-202, XV-202	1	86.0	40.0	32.54	158,861	174,747 (normal × 1.10)	A: 88 bar g / 120 °C, CS Cl 600	288.9	20.69	22.76	16,851	0.386	outside: v, ρv²	DN350
DN750-PG-202	PID-002	Converter feed (C-201 discharge): C-201 discharge → E-202 cold side inlet	3	80.6	48.6	20.05	565,272	621,799 (normal × 1.10)	B: 100 bar g / 120 °C, CS Cl 900	696.0	20.58	22.64	10,279	0.083	outside: v	DN800

Str. = HMB stream (6/b2 = cold shot to bed 2, its share of stream 6). Rated flow is evaluated at the governing case's density. Status lists the criteria not met at the rated flow. "Meets criteria at" = the smallest DN in the ID table that meets all three criteria.

3.1 Purge downstream of PV-203

DN150-PU-203 continues past PV-203 to the reformer fuel header. At an assumed header pressure of 4 bar g (not in the model) the rated purge has a density of 1.12 kg/m³ and would run at about 557 m/s in DN150 (Mach 0.72, sonic velocity 778 m/s), ρv² 347,980 kg/(m·s²) and 18.1 bar/100 m. The low-pressure criteria are met from DN700. The line number should break at PV-203 (EQ-ep-02).

4 Velocity in every case

Line no.	ID mm	Design kg/h	Turndown kg/h	End of run kg/h	Max rate kg/h	Design m/s	Turndown m/s	End of run m/s	Max rate m/s	Max ρv²
DN750-PG-201	696.0	565,272	422,108	630,362	622,795	23.93	19.21	23.66	25.85	11,754
DN300-PG-222	288.9	71,674	46,435	40,472	74,687	17.61	12.27	8.82	17.99	5,694
DN350-PG-232	317.5	115,293	78,108	99,075	127,714	23.45	17.08	17.87	25.47	11,414
DN350-PG-242	317.5	110,470	67,980	109,874	124,602	22.47	14.87	19.82	24.85	10,865
DN600-PG-213	548.1	267,836	229,586	380,941	295,792	24.37	22.87	31.72	26.32	14,225
DN900-PG-251	828.0	565,264	422,102	630,355	622,786	23.37	18.89	24.20	25.36	8,147
DN900-PG-205	844.0	433,748	327,541	509,376	479,043	12.65	10.20	12.79	13.80	3,283
DN150-PU-203	146.4	27,337	16,671	37,875	30,926	26.50	17.26	31.60	29.62	19,748
DN200-CM-209	193.7	131,516	94,561	120,979	143,743	1.52	1.09	1.40	1.66	2,255
DN300-SG-202	288.9	158,861	111,238	158,861	174,678	20.69	14.49	20.69	22.75	16,838
DN750-PG-202	696.0	565,272	422,108	630,362	622,795	20.58	16.62	20.56	22.29	10,135

Each case at its own temperature, pressure and density. The end-of-run case puts more gas through bed 1 (the quench valves close as the beds cool), which is why DN600-PG-213 is governed by it.

5 Process lines without a line number on the P&IDs

Sheet	Line	HMB stream	P bar g	T °C	ρ kg/m³	Normal kg/h	Rated kg/h (governing)	Design (EQL zone)	Suggested DN	v m/s	ρv²	ΔP bar/100 m
MEOH-PID-001	Bed-1 feed branch, quench header to E-201 cold side inlet	5	79.7	96.8	17.25	267,836	380,941 (End of run)	B: 100 / 170 °C, CS	DN650	19.11	7,106	0.068
MEOH-PID-001	TV-211 cold-side bypass round E-201	5	79.7	96.8	17.25	57,481	63,229 (normal × 1.10)	B: 100 / 170 °C, CS	DN300	15.53	4,162	0.097
MEOH-PID-001	E-201 cold side outlet to E-204 (bypass rejoins before E-204)	5	79.7	253.7	12.11	210,355	334,268 (End of run)	B: 100 / 335 °C, 1¼Cr-½Mo	DN750	18.23	4,527	0.037
MEOH-PID-001 / -002	E-201 hot side outlet to E-202 hot side	8	77.0	206.4	14.00	565,264	621,790 (normal × 1.10)	B: 100 / 280 °C, 1¼Cr-½Mo	DN1000	18.48	4,780	0.028
MEOH-PID-002	E-202 hot side outlet to E-203	9	76.7	155.9	15.59	565,264	621,790 (normal × 1.10)	B: 100 / 225 °C, 1¼Cr-½Mo	DN1000	16.59	4,293	0.025
MEOH-PID-002	C-201 suction after the make-up tie-in (streams 1 + 2)	3	75.0	40.0	19.19	565,272	621,799 (normal × 1.10)	A: 88 / 120 °C, CS	DN900	16.09	4,967	0.032
MEOH-PID-002	E-203 outlet to V-201 inlet (two-phase, 40 °C)	9							two-phase: size by piping on the mixed density (not sized here)			
MEOH-PID-002	PSV-201 inlet / outlet and HV-205 blowdown to flare	—							relief / blowdown: sized in MEOH-V1-PE-RLB-001			

These lines need numbers before they can be listed (EQ-ep-06). The suggested DN uses the same criteria and pipe walls as section 3.

6 Worked example · DN750-PG-201 (converter feed after E-202)

Step	Value	How
Normal flow	1,928.0 kNm³/h = 86,018 kmol/h = 565,272 kg/h	HMB stream 4
Gas molecular weight	6.572 kg/kmol	Σ y_i · MW_i
Density	17.25 kg/m³	ρ = P·MW/(R·T) = (79.70 + 1.013)·10⁵ · 6.572 / (8314 · 369.95)
Inside diameter	696.0 mm (A = 0.3805 m²)	OD 762 − 2 × 33.0 mm wall: zone B 100 bar g / 170 °C, CS A106 Gr B (section 2.1)
Rated flow	621,799 kg/h → 36,046 m³/h actual	governing: normal × 1.10
Velocity	26.32 m/s	Q / A
ρv²	11,948 kg/(m·s²)	against 15 000 (typical, assumed)
Reynolds number	2.16e+07	ρ·v·D/μ, μ = 0.0146 cP (assumed)
Darcy friction factor	0.01124	Churchill (1977), ε = 0.046 mm
ΔP per 100 m	0.097 bar	f · (100/D) · ρv²/2
Result	outside the 20 m/s basis; ρv² and ΔP inside	DN900 meets all three

7 Engineering queries raised by this document

No.	To	Issue	Proposed	Status
EQ-ep-01	MEOH-PID-001 / MEOH-PID-002	At the rated flow (the larger of normal × 1.10 and the highest-flow case) 7 of the 10 numbered gas lines exceed the 20 m/s basis of P&ID note 2 with the pipe walls for the design pressure zones of MEOH-V1-ME-EQL-001: DN750-PG-201 26.32 m/s (needs DN900); DN350-PG-232 25.79 m/s (needs DN400); DN350-PG-242 24.85 m/s (needs DN400); DN600-PG-213 31.72 m/s (needs DN800); DN900-PG-251 25.7 m/s (needs DN1050); DN300-SG-202 22.76 m/s (needs DN350); DN750-PG-202 22.64 m/s (needs DN800). ρv² and ΔP per 100 m are within the typical criteria on all of them except DN300-SG-202 (ρv² 16851).	Either increase these lines to the sizes listed (with piping to confirm wall and ID), or revise P&ID note 2 to a ρv²-based criterion for the high-pressure loop gas (typical ρv² ≤ 15 000 kg/(m·s²)) and keep the drawn sizes where ρv² is met; check the loop ΔP budget against C-201 either way.	open
EQ-ep-02	MEOH-PID-002	DN150-PU-203 carries the purge (105.1 kNm³/h normal, 37875 kg/h rated). Upstream of PV-203 at 75 bar g it runs at 31.6 m/s with ρv² 19748 kg/(m·s²) and 1.027 bar/100 m, outside all three typical criteria (DN200 meets them); the line number continues downstream of PV-203 to the reformer fuel header, where at an assumed 4 bar g the same flow would run at about 557 m/s (Mach 0.72). The fuel-header pressure is not in the model or on the P&ID.	Increase the upstream section to DN200; break the line number at the PV-203 spec break; size the downstream line for the fuel-header pressure (DN700 at 4 bar g with the typical 30 m/s limit) and confirm the header pressure; check PV-203 for noise.	open
EQ-ep-06	MEOH-PID-001 / MEOH-PID-002	8 process lines on the P&IDs have no line number (e.g. the E-201 → E-202 → E-203 → V-201 effluent train, the C-201 suction after the make-up tie-in, the TV-211 bypass, the PSV-201 and HV-205 flare lines), so the line sizing list cannot cover them.	Assign line numbers (and sizes) on the next P&ID revision; size them from the HMB streams given in MEOH-V1-PE-LSL-001 section 5.	open

8 Assumptions

1. Gas density from the ideal-gas law (Z = 1), as the model computes the loop pressure; a real H2-rich gas at 80 bar has Z ≈ 1.03–1.05, which lowers ρ and raises v by the same few per cent.

2. Gas viscosity 0.0130 cP at 40 °C × (T/313 K)^0.7; crude methanol 0.55 cP at 40 °C (assumed; not in the model).

3. Liquid densities 790 kg/m³ methanol and 995 kg/m³ water, volume-additive (the model's V-201 level calculation).

4. Latent heats 37.4 kJ/mol methanol and 43.3 kJ/mol water at 40 °C for the E-203 condensing duty (assumed; the model's E-203 heat is sensible only).

5. Line design pressure and temperature from MEOH-V1-ME-EQL-001: zone A 88 bar g (PSV-201 set), zone B 100 bar g; design temperature of the equipment each line connects. Pipe material carbon steel A106 Gr B (S = 138 MPa ≤ 204 °C, 130 MPa to 260 °C) up to the EQL carbon-steel limit in hot hydrogen (220 °C), 1¼Cr-½Mo A335 P11 (S = 125 MPa) above it; flanges Class 600 zone A, Class 900 zone B (all assumed).

6. Pipe walls: ASME B36.10 Sch 80 where listed and sufficient, else the ASME B31.3 304.1.2 minimum at the line design pressure, E = 1, Y = 0.4, + 1.5 mm corrosion allowance, ÷ 0.875 mill tolerance, rounded up to 1 mm (assumed; piping class to confirm).

7. Roughness 0.046 mm (commercial steel), Darcy friction factor from the Churchill (1977) equation.

8. Rated (design) flow for lines = the larger of normal × 1.10 and the highest flow of the four cases (typical, assumed).

9. Reformer fuel-header pressure 4 bar g (assumed, not in the model).

10. Margins: 10 % on flow, power and duty, 20 % on the purge (typical practice, assumed).